

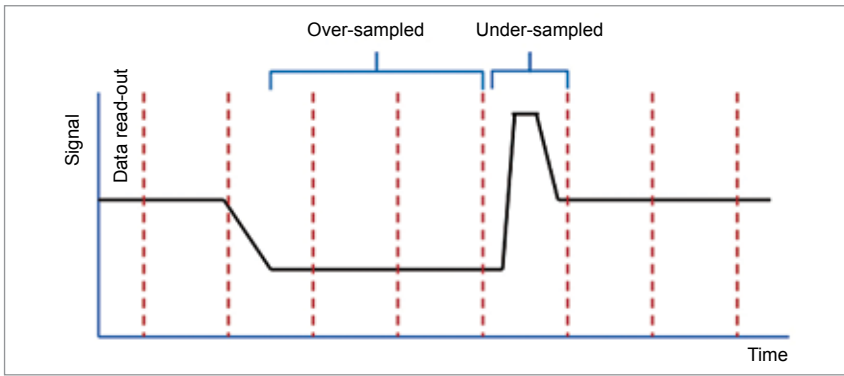


Fig. 6: Conventional image sensor behaviour (Credit: IDTechEx)

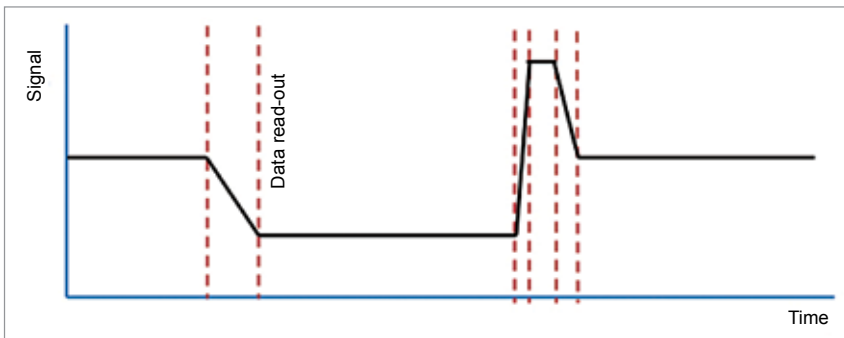


Fig. 7: Image based sensing behaviour (Credit: IDTechEx)

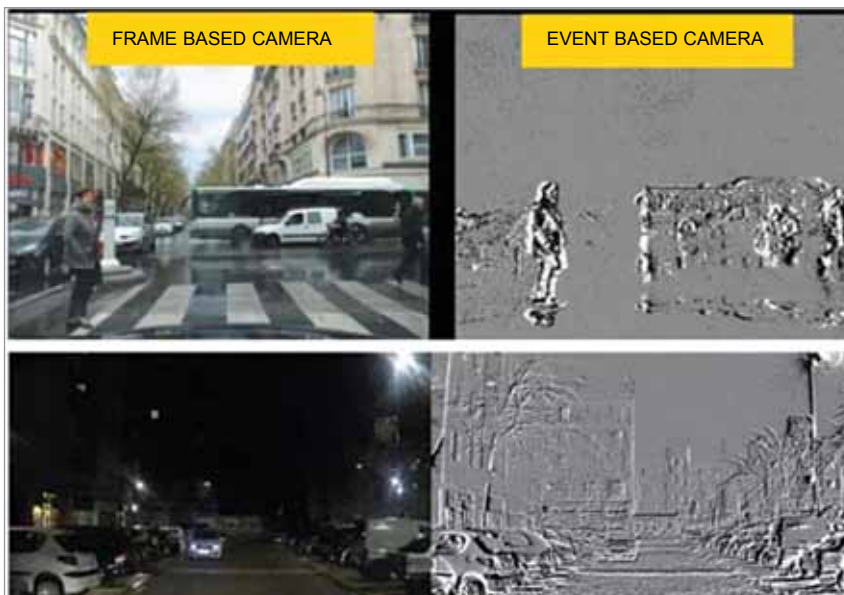


Fig. 8: Images from frame based and image based cameras (Credit: Prophesee)

scanning via drones. The snapshot has been recently commercialised and is used in astronomy.

Other applications of hyperspectral imaging include:

- Agriculture
- Infrastructure
- Oceanography
- Surveying and extraction
- Chemical identification
- Identification of food degradation products
- Environmental monitoring
- Defence

Event based imaging

It is a relatively new approach to data acquisition. Event based imaging, also known as dynamic vision sensing (DVS), only records changes that occur rather than the entire frame at specified intervals.

A conventional image sensor in a camera reads the intensity of each pixel at a pre-defined frame rate (for example, 25fps). Here, the data read-outs are equally spaced in time at a constant frequency.

Comparatively, in event based imaging, each pixel is read independently when the received intensity changes. So, rather than having data readouts at a constant frequency, you receive data having different frequency time-stamps based on intensity change.

To understand the advantage of image based sensing technology, let's take the help of graphs shown in Fig. 6 and Fig. 7, which depict a clear blue sky being recorded by an HD camera.

When using conventional sensing techniques, for most of the time, a camera records scenes where no relevant change occurs in the sky (see over-sampled in Fig. 6). And when a change does occur for a brief period, say an aeroplane passes by, the camera records that along with additional scenes of no change (see under-sampled in Fig. 6). All this happens because the data readouts have a fixed frequency.

Now, the same scenario recorded using image based sensing technology would focus on capturing only the relevant scenes related to the flying aeroplane (see Fig. 7). Since the data readouts have a flexible frequency

SWIR sensors. Hyperspectral imaging can be used outside the visible spectrum.

The basics of hyperspectral imaging technology lie in the availability of the entire spectrum wavelength at every pixel. So, while in a conventional image each pixel has red, green, and

blue values, in hyperspectral imaging an entire spectrum is present. This can be achieved through point scan, line scan, wavelength scan, and snapshot, with each having a high spectral resolution.

The line scan approach is very well suited for industrial imaging or land